

K1 CT to PT

K: I: 1, L: 1, N: 1 (I N L)
R: D: 1, U: 3 (U D U U)
Y: E: 4, L: 1 (E E E E L)
P: E: 1 (E)
T: F: 2, N: 1 (F N F)
S: H: 2 (H H)
A: T: 1 (T)
B: T: 1 (T)
D: S: 1, A: 2, N: 1 (A A S N)
E: A: 1, B: 1 (B A)
F: O: 1, S: 1, B: 1, W: 1 (W B S O)
G: A: 1 (A)
H: E: 2 (E E)
I: E: 1 (E)
J: C: 2, D: 1 (D C C)
L: S: 1, E: 1, G: 1, H: 1 (S G H E)
M: E: 1, I: 2 (E I I)
N: T: 1, B: 1 (T B)
Q: O: 2, H: 1, L: 2 (O L H L O)
U: T: 1, S: 2, I: 1 (T S I S)
V: T: 1, I: 1, N: 2 (N I T N)
X: G: 1, L: 1 (L G)
Z: N: 2 (N N)

K1 PT to CT

T: A: 1, B: 1, N: 1, U: 1, V: 1 (U A N B V)
O: F: 1, Q: 2 (Q Q F)
S: D: 1, F: 1, L: 1, U: 2 (L U F U D)
A: D: 2, E: 1, G: 1 (D D G E)
B: E: 1, F: 1, N: 1 (E F N)
C: J: 2 (J J)
D: R: 1, J: 1 (J R)
E: Y: 4, P: 1, H: 2, I: 1, L: 1, M: 1 (M P H Y H I Y Y L Y)
F: T: 2 (T T)
G: L: 1, X: 1 (L X)
H: S: 2, L: 1, Q: 1 (S S Q L)
I: K: 1, M: 2, U: 1, V: 1 (K U V M M)
L: K: 1, Y: 1, Q: 2, X: 1 (X Q Q K Y)
N: K: 1, T: 1, D: 1, V: 2, Z: 2 (Z Z K V T V D)
U: R: 3 (R R R)
W: F: 1 (F)

P: EADVQVM
A: MXKJVJF
L: UYRYYYD
I: FUNQUQ
M: PSSTVT
P: HDHQLM
S: ZJGULK
E: LKNXTY
S: RZFQRR
T: FLIBED

P: BTANLNI
A: ELNCICO
L: TEDEEEN
I: WSTOSO
M: EHHFTF
P: EAELHI
S: NDAIEL
E: SIBGNL
S: UNSHUU
T: BGETAS

K2 CT TO PT

K: E: 7, L: 2, N: 6 (L E L E N N N N E N E E N E E N)

R: M: 1, N: 6, U: 4 (N N U N U U M N U N N)

Y: G: 1, N: 3, U: 1, V: 1 (V N N U N G)

P: H: 5, U: 3, V: 3, W: 4 (W U H W U W H H H V V V H U W)

T: I: 4, U: 2, W: 5, X: 2 (I W I U U W W I X W I X W)

O: W: 2, X: 1 (W W X)

S: K: 2, W: 1, X: 2 (X K K W X)

A: R: 6, M: 1, X: 2 (R X M R X R R R R)

B: R: 2, Y: 2, N: 3 (N R N Y Y N R)

C: R: 4, Y: 2, P: 1 (P Y R R R Y R)

D: R: 3, Y: 2, T: 17, U: 2 (T T T R T T T U T T T T T R T T T T Y T R T T Y U)

E: Y: 1, P: 1, T: 13, O: 13, V: 1 (O T Y O O O T T T T T O O O T T O T O T T V P O O O O T T T)

F: T: 2, O: 7, S: 17, W: 1 (T T S S O O O S O S O S S S S O O W S S S S S S S S S S)

G: T: 6, O: 3, S: 8, A: 6 (S S S S T T A A A A O A T S A T T S T S O S O)

H: O: 3, S: 2, A: 6, B: 3 (A B B A A O A B O S A S A O)

I: K: 1, S: 1, A: 2, B: 1, C: 2 (A C K A B C S)

J: R: 1, A: 4, C: 1, D: 7 (A A D A D D D D R D D C A)

L: Y: 3, C: 1, D: 2, E: 12 (E E D E Y E E E E C E E D Y E E E Y E)

M: C: 1, E: 10, F: 5 (E E E F E C E E F F E E E F F E)

N: T: 3, D: 2, E: 8, F: 1, G: 1 (T T G E E D E E E F E T E E D)

Q: O: 1, E: 10, F: 1, G: 4, H: 8 (H H H E H G H F E G E E H E O H G G E E E H E E)

U: S: 6, G: 1, H: 5, I: 8 (S S H S I I G H H I H H I I I S I I S S)

V: A: 2, G: 2, H: 1, I: 11 (I I I I I A I H A I G I I I G I)

W: H: 1, I: 2, L: 5 (L I I L L H L L)

X: C: 1, I: 1, L: 2, M: 3 (M M L L I C M)

Z: D: 3, L: 4, M: 2, N: 9 (L N D L D N N L N D M N L N M N N N)

A: VHGGGCMQMKFUEEQHKKKTVFYEMUUFKGGHQCQEVOPGPMGKEJEG

B: FZWFWPIRZQRNCNCRZSUIHPCVOGNLZOHNCDRENRRNRPDGW

S: PWHQHfALJEPAGJRREJWHDWULQSLGBGDTULFREFEZJUDMZJ

C: JEKJKQGGADJQYTTYMHQHWNQDQYAGSIRGVQQGXQDGQVQHNL

I: UTKNKZPTNAUZUBBIVKLDKTZKJQVTFPKPPUEKBXNLYPUDUL

S: DZQCDDFIQGNGXJJZDFSDBDEFLPIEDUFUDMLFDEQFADMAPA

S: EYEEQQXMLDGGZULDEUWZDFFRETFDZQFFAJUZFMZFLLXEFLE

A: EVTGMMHVVELEBFTHFUUIEZTEXFVXHEMNZVVLMPMVBGMT

K2 PT TO CT
K: S: 2, I: 1 (I S S)
R: A: 6, B: 2, C: 4, D: 3, J: 1 (A D A C B C C D J A C D A B A A)
Y: B: 2, C: 2, D: 2, E: 1, L: 3 (E C L B B C D L D L)
P: C: 1, E: 1 (C E)
T: D: 17, E: 13, F: 2, G: 6, N: 3 (F D E F N D D G G N D E E E E D D D D D D E E E D G D D G E D E G D N G D D E E E)
O: E: 13, F: 7, G: 3, H: 3, Q: 1 (E E E E F F E E F H F H F E E G Q F E F E E E E H G G)
S: F: 17, G: 8, H: 2, I: 1, U: 6 (U G G G G F F U U F H F F F F G I F H F F U F F F F G F F G U F U G)
A: G: 6, H: 6, I: 2, J: 4, V: 2 (J H J I H G J G G G H V H I V G G H H J)
B: H: 3, I: 1 (H H H I)
C: I: 2, J: 1, L: 1, M: 1, X: 1 (I M I L X J)
D: J: 7, L: 2, N: 2, Z: 3 (Z J Z L J J J J N Z J L J N)
E: K: 7, L: 12, M: 10, N: 8, Q: 10 (K K M Q M M L M L N Q N L K Q K Q L K M Q L N L K N N M L Q Q L K N Q M N L M Q L M N Q M L L)
F: M: 5, N: 1, Q: 1 (M Q M N M M M)
G: Y: 1, N: 1, Q: 4, U: 1, V: 2 (Q N Q U Q V Q Y V)
H: P: 5, Q: 8, U: 5, V: 1, W: 1 (Q Q Q Q P Q U U W U V Q U U Q P P P Q P)
I: T: 4, U: 8, V: 11, W: 2, X: 1 (V T V W W T V V V U U V T U T V U U U U V V X V U V)
L: K: 2, W: 5, X: 2, Z: 4 (Z W K K Z Z W W W X Z X W)
M: R: 1, A: 1, X: 3, Z: 2 (X A X Z R Z X)
N: K: 6, R: 6, Y: 3, B: 3, Z: 9 (Z R K R Z Y B B R Y Z K K K Z Y Z K Z R B Z R Z R K Z)
U: R: 4, Y: 1, P: 3, T: 2, D: 2 (P T R T R D P Y R R P D)
V: Y: 1, P: 3, E: 1 (Y E P P P)
W: P: 4, T: 5, O: 2, S: 1, F: 1 (P T P T T O T P S F O P T)
X: T: 2, O: 1, S: 2, A: 2 (A S T A T O S)

A: IASSSYEGENOHTTGANNNWIOUTEHHONSAGYGTIXVSVESNTCTW
B: TLITIUANLNFNERERNLXGASURHWOECLWSEERYNPENENEUYOXO
S: WLBHBSREDOWRADUUODLBTIEHKEAYATXIESUOSONDITFNL
C: AYLALETTXRAENUUNCOEHOHDERENXTWSMTGEETIERTEGEODA
I: SIETEDHITMSDSNNKAEYUEIDEROAIWHEHHSVENCTYGHSUSY
S: TNHPTTSCHAGAMDDNTSKTYTTOSEWCOTISITFESTOHSRTFRWE
S: OVOOHMFETANIETOILNTSSUOXSTNHSSRDINSFNSEEMOSER
A: TIWSEEAIITDTROWOAOHHBTMWTLOILATEFMIIDEVEIREST

Python Script

```
def count_transformations(decoded, encoded):
    transformation_dict = {}

    for d, e in zip(decoded, encoded):
        if d not in transformation_dict:
            transformation_dict[d] = {}
        if e not in transformation_dict[d]:
            transformation_dict[d][e] = 0
        transformation_dict[d][e] += 1

    return transformation_dict

def sort_transformations(transformations, custom_alphabet):
    sorted_transformations = {}
    for d in custom_alphabet:
        if d in transformations:
            sorted_transformations[d] = dict(sorted(transformations[d].items(), key=lambda x: custom_alphabet.index(x[0])))

    return sorted_transformations

def print_transformations(decoded, encoded, sorted_transformations):
    for d, encoded_letters in sorted_transformations.items():
        transformations_str = ', '.join([f"{e}: {count}" for e, count in encoded_letters.items()])
        occurrences = [e for d_char, e in zip(decoded, encoded) if d_char == d]
        print(f"{d}: {transformations_str} ( { ' '.join(occurrences)} )")

string1 =
"VFPJUDEEHZWETZYVGWHKKQETGFQJNCEGGWHKKDQMCPFQZDQMMIAGPFXHQRLGTIMVMZJANQLVKQEDAGDVFRPJUNGEUNAQQZGZLECGYUXUEENJTBJLBQCRTBJDFHRRYIZETKZEMVD
UFKSJHKFWHKUWQLSZFTIHDDDUVHDWKBFUFPWNTDFIYCUQZEREVLDKFEZMOQQJLTTUGSYQPFEUNLAVIDXFLGGTEZFKZBSFDQVGOGIPUFXXHHRKFFHQNTGPUAECNUVPDJMQCLQU
MUNEDFQELZZVRRGKFFVOEEXBDMVPNFQXEZLGREDNQFMPNZGLFLPMRJQYALMGNUVPDXVKPDQUMEBEDMHDAFMJGZNUPLGESWJLLAETG"
string2 =
"ITWASTOTALLYINVISIBLEHOWSTHATPOSSIBLETHEYUSEDTHEEARTHSMAGNETICFIELDXTHEINFORMATIONWASGATHEREDANDTRANSMITTEDUNDERGRUUNDTOANUNKNOWNLOCATIONXD
OESLANGLEYKNOWABOUTTHISTHEYSHOULDTISBURIEDOUTTHERESOMEWHEREXWHOKNOWSTHEEXACTLOCATIONONLYWWTHISWASHISLASTMESSAGEXTHIRTYEIGHTDEGREESFIFTYSEV
ENMINUTESEXPOINTFIVESECONDSNORTHSEVENTYSEVENDEGREESEIGHTMINUTESFORTYFOURSECONDSWESTXLAYERTWO"

custom_alphabet = "KRYPTOSABCDEFGHIJLMNQUVWXZ"

transformations = count_transformations(string1, string2)
sorted_transformations = sort_transformations(transformations, custom_alphabet)
print_transformations(string1, string2, sorted_transformations)
```